

1. Mathematical Induction

- Axiom (Well-ordering Principle of $\mathbb{N}$)** - Every non-empty subset of $\mathbb{N}$ has a minimal element.
- Thm (PMI)** - Suppose $S \subseteq \mathbb{N}$ satisfies: (1) $1 \in S$; (2) If $k \in S$, then $k+1 \in S$. Then $S = \mathbb{N}$.
- Thm (PMI in practice)** - Suppose (1) $P(1)$ holds; (2) $P(k) \Rightarrow P(k+1)$. Then $P(n)$ holds for all $n \in \mathbb{N}$.
- Thm (Strong induction)** - Suppose (1) $P(1)$ holds; (2) $P(k)$ holds for all $k' \leq k$ implies $P(k+1)$ holds. Then $P(n)$ holds for all $n \in \mathbb{N}$.

2. Countability of Infinite Sets

- Thm** - If $S_1, \dots, S_n$ are finite, then $S = \bigcup_{j=1}^n S_j$ is finite, and $|S| \leq \sum_{i=1}^n |S_i|$, where equality holds if and only if S_i are pairwise disjoint.
- Defn** - An infinite set S is **countable** if there exists a bijection $f: \mathbb{N} \rightarrow S$.
- Thm*** - Let S be infinite and $S = \bigcup_{i=1}^{\infty} F_i$, where F_i are finite sets. Then S is countable.
- Cor** - If $S_1, \dots, S_n$ are countable, then $S = \bigcup_{j=1}^n S_j$ is countable.
- Cor** - If $S_1, S_2, \dots$ are countable, then $S = \bigcup_{j=1}^{\infty} S_j$ is countable.
- Thm (Schröder-Bernstein)** - If there exist injections $f: A \rightarrow B$ and $g: B \rightarrow A$, then there exists a bijection $h: A \rightarrow B$.
- Prop** - $\mathcal{P}(\mathbb{N})$ is uncountable.

3. Inequalities

- Bernoulli's inequality** - For $x > -1, n \in \mathbb{N}$, we have $(1+x)^n \geq 1+nx$.

- AM-GM inequality** - For $a_1, \dots, a_n \geq 0$,
$$\frac{a_1 + \dots + a_n}{n} \geq \sqrt[n]{a_1 \dots a_n}$$

where equality holds iff $a_1 = \dots = a_n$.

- GM-HM inequality** - For $a_1, \dots, a_n > 0$,
$$\sqrt[n]{a_1 \dots a_n} \geq \frac{n}{\frac{1}{a_1} + \dots + \frac{1}{a_n}}$$

where equality holds iff $a_1 = \dots = a_n$.

- Triangle inequality** - For $a, b \in \mathbb{R}$,
$$||a| - |b|| \leq |a+b| \leq |a| + |b|.$$

For $a_1, \dots, a_n \in \mathbb{R}$,

$$|a_1 + \dots + a_n| \leq |a_1| + \dots + |a_n|.$$

- Cauchy-Schwarz inequality** - For $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$,

$$\left(\sum_{i=1}^n a_i b_i \right)^2 \leq \left(\sum_{i=1}^n a_i^2 \right) \left(\sum_{i=1}^n b_i^2 \right)$$

where equality holds iff $a_i = k b_i$ for some $k \in \mathbb{R}$.

4. Completeness of Real Numbers

Supremum property

- Defn** - $\beta = \sup E$ is **supremum** of $E \subseteq S$ if

- β is an upper bound of E ;
- for any $\beta' < \beta$, β' is not an upper bound of E .

- Defn** - $\beta = \inf E$ is **supremum** of $E \subseteq S$ if

- β is a lower bound of E ;
- for any $\beta' < \beta$, β' is not a lower bound of E .

- Ordered set S has the **supremum property** if every subset bounded above has a supremum (in S).

Dedekind cuts

- Defn** - (A, A') is **Dedekind cut** if

- $A, A' \neq \emptyset$;
- $A \cap A' = \emptyset, A \cup A' = \mathbb{Q}$;
- For any $a \in A, a' \in A', a < a'$;
- A does not admit a maximal element, namely, for any $a \in A$, there exists $b \in A$ such that $b > a$.

- Defn** - $\mathbb{R} := \{ (A, A') : (A, A') \text{ is a Dedekind cut} \}$

- Defn** - Order in $\mathbb{R}$ is defined as: $A < B$ if $A \subset B$; $A > B$ if $B \subset A$.

- Prop** - Order on $\mathbb{R}$ is a total order: for any $A, B \in \mathbb{R}$, exactly one of $A = B, A < B, A > B$ holds.

- Thm** - $\mathbb{R}$ has the supremum property.

Cor - Infimum property

- Addition** - For $m_1 = (M_1, M'_1)$ and $m_2 = (M_2, M'_2)$ in $\mathbb{R}$, define $m_1 + m_2 = (M_3, M'_3)$, where $M_3 = M_1 + M_2 := \{a_1 + a_2 : a_1 \in M_1, a_2 \in M_2\}$.

- Inverse** - For $m = (M, M')$, the inverse of m is defined as follows: If $m \in \mathbb{Q}$, then $-m$ is defined as in $\mathbb{Q}$. If $m \notin \mathbb{Q}$, then

$$-m = (-M', -M)$$

where $-M' := \{-m' : m' \in M'\}$.

- Product** - For $m_1 = (M_1, M'_1)$ and $m_2 = (M_2, M'_2)$ in $\mathbb{R}$, define

$$m_1 \cdot m_2 = m_3 = (M_3, M'_3)$$

as follows:

- If $m_1 = 0$ or $m_2 = 0$, then $m_3 = 0$.
- If $m_1 > 0$ and $m_2 > 0$, then

$$M_3 = \{xy : x, y \geq 0, x \in M_1, y \in M_2\} \cup \{x : x < 0\}.$$

- If $m_1 > 0$ and $m_2 < 0$, then $m_3 = -(m_1 \cdot (-m_2))$.
- If $m_1 < 0$ and $m_2 > 0$, then $m_3 = -((-m_1) \cdot m_2)$.
- If $m_1 < 0$ and $m_2 < 0$, then $m_3 = (-m_1) \cdot (-m_2)$.

- Reciprocal** - For $m = (M, M') \neq 0$, define $\frac{1}{m} = (B, B')$ as follows:

- If $m > 0$, then $B' = \{\frac{1}{x} : x > 0, x \in M\}$.
- If $m < 0$, then $\frac{1}{m} = -\frac{1}{m}$.

Archimedean property

- Archimedean property** - For any $x, y > 0$, there exists $n \in \mathbb{N}$ such that $nx > y$.

- Cor** - For any $c \in \mathbb{R}$, there exists a unique integer $[c] \in \mathbb{Z}$ such that $[c] \leq c \leq [c] + 1$.

- Cor** - For any $x > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$.

- Cor** - $\mathbb{Q}$ is dense in $\mathbb{R}$: for any $a < b$, there exists $r \in \mathbb{Q}$ such that $a < r < b$.

Nested intervals

- Nested interval property** - Let $\{I_n = [a_n, b_n] : n \in \mathbb{N}\}$ be nested. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

- Tut** - Let $\{I_n = [a_n, b_n] : n \in \mathbb{N}\}$ be a sequence of nested, $|I_n| < 1/n$ (i.e., $|I_n| \rightarrow 0$). Then $\bigcap_{n=1}^{\infty} I_n = \{\xi\}$ is a singleton.

- Thm** - $\mathbb{R}$ is uncountable.

Proof: Contradiction. Construct a sequence of nested intervals such that each interval avoids one more point of $\mathbb{R}$ than the previous. Then apply nested interval property.

5. Limits of Sequences and Series

Limits of sequences

- Defn** - $\{a_n\}$ **converges** to A if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad |a_n - A| < \varepsilon.$$

That is, all but finitely terms eventually lie in an ε -neighbourhood centered at A .

In proofs, we try to bound the quantity $|a_n - A|$.

- Prop** - $\lim_{n \rightarrow \infty} a_n = A \iff \lim_{n \rightarrow \infty} (a_n - A) = 0 \iff \lim_{n \rightarrow \infty} |a_n - A| = 0$

Properties of limits

- Prop** - Convergent $\Rightarrow$ bounded

Proof: Bound terms $n \geq N$, then bound finitely many terms $|a_1|, \dots, |a_{N-1}|$.

- Lemma** - Suppose $\{a_n\}$ is convergent, $a_n \neq 0$, $\lim_{n \rightarrow \infty} a_n \neq 0$. Then there exists $C > 0$ such that $|a_n| \geq C$ for all $n \in \mathbb{N}$.

- Thm (Order preserving)** - If $\{a_n\}, \{b_n\}$ convergent, $a_n \geq b_n$, then $\lim_{n \rightarrow \infty} a_n \geq \lim_{n \rightarrow \infty} b_n$.

- Limit laws** - Suppose $\lim_{n \rightarrow \infty} a_n = a, \lim_{n \rightarrow \infty} b_n = b$. Then

$$a_n \pm b_n \rightarrow a \pm b, \quad \lambda a_n = \lambda a, \quad a_n b_n \rightarrow ab, \quad \frac{a_n}{b_n} \rightarrow \frac{a}{b}.$$

To evaluate limit of rational functions $\lim_{n \rightarrow \infty} \frac{P(n)}{Q(n)}$ where P, Q are polynomials, divide by the highest power.

- Squeeze theorem** - If $a_n \leq b_n \leq c_n$, and $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = A$, then $\lim_{n \rightarrow \infty} b_n = A$.

- Let $a_n \geq 0, a_n \rightarrow a$. Then $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}$.

- $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$.

Monotone convergence theorem

- Monotone convergence theorem** - If $\{a_n\}$ increasing [decreasing] and bounded from above [below], then $\{a_n\}$ is convergent.
- For recursively defined sequences, use induction to show (1) increasing/decreasing; (2) bounded above/below.

- Defn** - **Euler's number** $e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$.

- $a_n = \left(1 + \frac{1}{n}\right)^n \nearrow e \quad b_n = \left(1 + \frac{1}{n}\right)^{n+1} \searrow e$
 $\{b_n\}$ converges since $b_n = a_n \left(1 + \frac{1}{n}\right)$ and $\{a_n\}$ converges.

Subsequences, Bolzano-Weierstrass theorem

- Defn** - Given a sequence $\{a_n\}$, we say that $\{b_k\}_{k \in \mathbb{N}}$ is a subsequence of $\{a_n\}$ if for any $k \in \mathbb{N}$,

$$b_k = a_{n_k}$$

where $n_1 < n_2 < \dots$.

- Thm** - If a sequence converges, then all subsequences converge to the same limit.

- Cor** - If $\{a_n\}$ has two subsequences with different limits, then $\{a_n\}$ is divergent.

- Monotone subsequence theorem** - Any sequence admits a monotone subsequence.

Proof: We call a_n a *peak* if $a_n \geq a_m$ for all $m \geq n$. Consider cases: infinitely/finitely many peaks, construct decreasing/increasing subsequence.

- Bolzano-Weierstrass theorem** - Any bounded sequence admits a convergent subsequence.

- Tut** - $\sum \sin n$ is divergent.

Proof: Divergence Test, construct a subsequence $\sin n_k \geq \frac{1}{2}$, so $\sin n_k \not\rightarrow 0$

Cauchy sequences

- Defn** - $\{a_n\}$ is a **Cauchy sequence** if

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m, n \geq N, \quad |a_m - a_n| < \varepsilon.$$

- Thm** - Convergent $\iff$ Cauchy

- Defn** - $\{a_n\}$ is **contractive** if there exists a constant $0 < C < 1$ such that $|a_{n+1} - a_n| \leq C|a_n - a_{n-1}|$ for all $n \geq 2$.

- Prop** - Contractive $\Rightarrow$ Cauchy $\Rightarrow$ convergent

- Alternating harmonic series** $\sum \frac{(-1)^{n-1}}{n}$ is convergent.

Properly divergent sequences

- Defn** - $\lim_{n \rightarrow \infty} a_n = +\infty$ if $\forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n > C$.

$$\lim_{n \rightarrow \infty} a_n = -\infty \text{ if } \forall C \in \mathbb{R}, \quad \exists N \in \mathbb{N}, \quad \forall n \geq N, \quad a_n < C.$$

$$\lim_{n \rightarrow \infty} a_n = \infty \text{ if } \{|a_n|\} \text{ tends to } +\infty.$$

- Squeeze theorem** - For $\{a_n\}$ and $\{b_n\}$,

$$\circ \text{ if } a_n \geq b_n \text{ and } \lim_{n \rightarrow \infty} b_n = +\infty, \text{ then } \lim_{n \rightarrow \infty} a_n = +\infty;$$

$$\circ \text{ if } a_n \geq b_n \text{ and } \lim_{n \rightarrow \infty} a_n = -\infty, \text{ then } \lim_{n \rightarrow \infty} b_n = -\infty.$$

- Thm (Reciprocals)** -

$$\circ \text{ Suppose } a_n \neq 0 \text{ and } \lim_{n \rightarrow \infty} a_n = \infty. \text{ Then } \lim_{n \rightarrow \infty} \frac{1}{a_n} = 0.$$

$$\circ \text{ Suppose } a_n \neq 0 \text{ and } \lim_{n \rightarrow \infty} a_n = 0. \text{ Then } \lim_{n \rightarrow \infty} \frac{1}{a_n} = \infty.$$

Upper and lower limits

- Defn** - **Upper limit** $\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup\{a_k : k \geq n\}$

- Defn** - **Lower limit** $\liminf_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \inf\{a_k : k \geq n\}$

- Defn** - A is a **limit point** of a sequence $\{a_n\}$ if A is the limit of some subsequence of $\{a_n\}$.

- Thm** - Let $\{a_n\}$ be bounded; let E denote the set of limit points of $\{a_n\}$. Then both upper limit and lower limit of $\{a_n\}$ are contained in E . Moreover,

$$\limsup_{n \rightarrow \infty} a_n = \max E, \quad \liminf_{n \rightarrow \infty} a_n = \min E.$$

- Cor** - For any bounded sequence $\{a_n\}$,

$$\liminf_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} a_n,$$

where equality holds if and only if $\{a_n\}$ is convergent.

- Prop (Order preserving)** - Let $\{a_n\}, \{b_n\}$ bounded, $a_n \leq b_n$. Then

$$\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} b_n, \quad \limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} b_n.$$

Infinite series

- **Defn** - Sequence of partial sums $s_n = \sum_{k=1}^n a_k$. If $s_n \rightarrow A$, denote $\sum a_n = A$.
- **Divergence test** - If $\sum a_n$ convergent, then $a_n \rightarrow 0$.
Contrapositive: if $a_n \not\rightarrow 0$, then $\sum a_n$ divergent.
- **Geometric series** $\sum r^n$ diverges if $r \geq 1$, converges if $0 < r < 1$.
- **Cauchy criterion** - $\sum a_n$ converges iff Cauchy:

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}, \quad \forall m > n \geq N, \quad \left| \sum_{k=n+1}^m a_k \right| < \varepsilon.$$

- **Monotone convergence theorem** - Suppose (1) $a_n \geq 0$; (2) partial sums of $\sum a_n$ are bounded. Then $\sum a_n$ is convergent.
- **Defn** - $\sum a_n$ is **absolutely convergent** if $\sum |a_n|$ convergent.
Defn - $\sum a_n$ is **conditionally convergent** if convergent but not absolutely convergent.
- **Prop** - Absolute convergent $\Rightarrow$ convergent

Various convergence tests

- **Comparison test** - Suppose $\{a_n\}$ and $\{b_n\}$ satisfy that there exist $K \in \mathbb{N}$ and $\lambda > 0$ such that for all $n \geq K$,
$$0 \leq a_n \leq \lambda b_n.$$
 - If $\sum b_n$ is convergent, then $\sum a_n$ is convergent.
 - If $\sum a_n$ is divergent, then $\sum b_n$ is divergent.
- For $0 < p \leq 1$, $\frac{1}{n^p} \geq \frac{1}{n} > 0$, so $\sum \frac{1}{n^p}$ is divergent.
For $p \geq 2$, $0 < \frac{1}{n^p} \leq \frac{1}{n^2}$, so $\sum \frac{1}{n^p}$ is convergent.
- **Limit comparison test** - Suppose $\{a_n\}$ and $\{b_n\}$ are both positive sequences and
$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lambda.$$
 - If $\lambda > 0$, then $\sum a_n$ convergent $\Leftrightarrow \sum b_n$ convergent.
 - If $\lambda = 0$, then $\sum b_n$ convergent $\Rightarrow \sum a_n$ convergent; $\sum a_n$ divergent $\Rightarrow \sum b_n$ divergent.

- **p-series** - For $p > 1$, $\sum \frac{1}{n^p}$ is convergent.
Proof: dyadic partition
- **Harmonic series** $\frac{1}{n}$ divergent, **2-series** $\sum \frac{1}{n^2}$ convergent
- **Cauchy condensation test** - Suppose $\{a_n\}$ is positive and decreasing. Then $\sum a_n$ is convergent if and only if

$$\sum_{n=1}^{\infty} 2^n a_{2^n}$$

- is convergent.
Note: useful to deal with denominator $\log_2 n$
- **Leibniz test / Alternating series test** - Suppose $a_n \geq 0$, $a_n \searrow 0$. Then $\sum (-1)^{n+1} a_n$ is convergent.
 - $\sum \frac{(-1)^n}{\sqrt{n}}$ convergent
 - **Root test** - Let $\{a_n\}$ be a sequence.

- If there exist $0 \leq r < 1$ and $K \in \mathbb{N}$ such that
$$|a_n|^{1/n} \leq r \quad \text{for all } n \geq K,$$
then $\sum a_n$ is absolutely convergent.
- If there exists $K \in \mathbb{N}$ such that
$$|a_n|^{1/n} \geq 1 \quad \text{for all } n \geq K,$$
then $\sum a_n$ is divergent.

- **Cor** - Suppose $\limsup_{n \rightarrow \infty} |a_n|^{1/n} = r \geq 0$.
 - If $r < 1$, then $\sum a_n$ absolutely convergent.
 - If $r > 1$, then $\sum a_n$ divergent.
- **Ratio test** - Let $\{a_n\}$ be a sequence with non-zero terms.

- If there exist $0 < r < 1$ and $K \in \mathbb{N}$ such that
$$\left| \frac{a_{n+1}}{a_n} \right| \leq r \quad \text{for all } n \geq K,$$
then $\sum a_n$ is absolutely convergent.
- If there exists $K \in \mathbb{N}$ such that
$$\left| \frac{a_{n+1}}{a_n} \right| \geq 1 \quad \text{for all } n \geq K,$$
then $\sum a_n$ is divergent.

- **Cor** - Suppose $a_n \neq 0$ and $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = r \geq 0$.
 - If $r < 1$, then $\sum a_n$ is absolutely convergent.
 - If $r > 1$, then $\sum a_n$ is divergent.
 - If $r = 1$, then the test is inconclusive.
- **Prop** - $\sum_{n=0}^{\infty} \frac{1}{n!} = e$.
- **Cor** - e is irrational.
Proof: Consider $q!(e - s_q)$
- **Tut** - $\lim_{n \rightarrow \infty} n(e^{1/n} - 1) = 1$
Proof: Squeeze $a_n \leq e \leq b_{n-1}$, then manipulate
- **Cauchy's first limit theorem** - If $\lim_{n \rightarrow \infty} a_n = a$, then

$$\lim_{n \rightarrow \infty} \frac{a_1 + \cdots + a_n}{n} = a.$$

- **Stolz theorem** - Let $\{a_n\}$ be a sequence, $\{b_n\}$ strictly increasing and divergent. If
$$\lim_{n \rightarrow \infty} \frac{a_{n+1} - a_n}{b_{n+1} - b_n} = A$$
then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = A$.
- **Tut** - Suppose $a_n > 0$, $\sum a_n$ divergent. Let $s_n = \sum_{k=1}^n a_k$. Then $\sum \frac{a_n}{s_n}$ divergent.

- **Tut** - Suppose $a_n > 0$, $\sum a_n$ convergent. Then $\sum \frac{\sqrt{a_n}}{n}$ convergent.
- **Tut** - Suppose $a_n > 0$, for any $b_n \searrow 0$, $\sum a_n b_n$ convergent. Then $\sum a_n$ convergent.
- **Tut (Feteke)** - Suppose $a_n > 0$, $a_{m+n} \leq a_m + a_n \ \forall m, n \in \mathbb{N}$. Then $\{\frac{a_n}{n}\}$ convergent.

Abel's summation formula

- **Abel's summation formula** - Let $A_n = \sum_{k=1}^n a_k$. For any $q > p$,

$$\sum_{n=p}^q a_n b_n = A_q b_q - A_{p-1} b_p + \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}).$$

- Note: useful in proofs to show $\sum a_n b_n$ is Cauchy.
- **Dirichlet's test** - Suppose (1) partial sums of $\sum a_n$ are bounded; (2) $b_n > 0$, $b_n \searrow 0$. Then $\sum a_n b_n$ is convergent.
 - **Abel's test** - Suppose (1) partial sums of $\sum a_n$ are convergent; (2) $\{b_n\}$ monotone and bounded. Then $\sum a_n b_n$ is convergent.

Rearrangement of series

- **Defn** - $\{b_n\}$ is a **rearrangement** of $\{a_n\}$ if $b_n = a_{f(n)}$, where $f: \mathbb{N} \rightarrow \mathbb{N}$ is bijection.
- **Thm** - Let $\{a_n\}$ be a sequence such that $\sum a_n$ is absolutely convergent. Let $\{b_n\}$ be a rearrangement of $\{a_n\}$. Then $\sum b_n$ is absolutely convergent, and $\sum b_n = \sum a_n$.
- **Riemann rearrangement theorem** - Let $\{a_n\}$ be a sequence such that $\sum a_n$ converges conditionally. Then for any $A \in \mathbb{R}$, there exists a rearrangement $\{b_n\}$ of $\{a_n\}$ such that $\sum b_n = A$.
- **Tut** - Suppose $a_n > 0$, $\sum a_n$ convergent. Let $r_n = \sum_{k=n}^{\infty} a_k$. Then $\sum \frac{a_n}{\sqrt{r_n}}$ convergent, $\sum \frac{a_n}{r_n}$ is divergent (not Cauchy).